Nanotechnology

Nanoscience

- It is the study of the fundamental principles of molecules and structures with atleast one dimension roughly between 1nm and 100nm.

Nanotechnology

- Nanotechnology is the principles of manipulation of atom by atom through control of the structure of matter at the molecular level.

In short, nanoscience is the study of nanostructures and nanotechnology is the application of these knowledge in different industries.

Significance of the Nanoscale

Most of the properties of a solid depends on size of the solid. For bulk materials properties like resistivity, density, elastic modulus etc are averaged properties. When size of the material becomes smaller and smaller this averaging no longer works and the properties of materials change drastically in nanometer range.

Two main factors that causes significant change in the properties of nanomaterials from their bulk counterparts is increased surface to volume ratio and quantum effects.

1. Surface to volume ratio

ex: For spherical nanoparticle

$$\frac{\text{Surface area}}{\text{Volume}} = \frac{4\pi R^2}{\frac{4}{3}\pi R^3} = \frac{3}{R}$$

Hence surface to volume ratio increases as radius R decreases since it is inversely proportional to radius.

As the particle size decreases a greater proportion of atoms are found at the surface compared to those inside. Thus nanoparticles have a much greater surface area ~~per su~~ when compared with larger particles. As surface area increases, properties like surface reactivity, catalytic activity, electrical and thermal conductivity etc change remarkably.

2. <u>Quantum confinement</u>

Due to quantum confinement the properties change significantly because energy levels become discrete and motion of electrons become restricted. Quantum effects begin to dominate the behaviour of matter at the nanoscale affecting optical, electrical and magnetic behaviour of materials.

Classification of Nanostructures

Based on the number of dimensions that are confined, nanostructures are classified as quantum well (nanosheet), quantum wire (nanowire) and quantum dots.

1) Nanasheets (quantum well)

In nanosheets confinement is present in only one dimension ie, along the thickness. Nanosheets are two dimensional structures where carriers are allowed to move freely along a two dimensional plane.

The wavefunction and energy in this case is

$$\psi_n(x,y,z) = \sqrt{\frac{2}{L_z}} \sin\left(\frac{n_z \pi z}{L_z}\right) e^{ik_x x} e^{ik_y y}$$

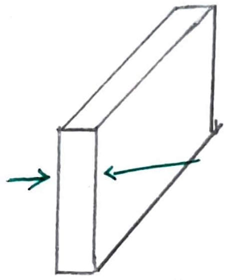
$$E_n = \frac{n_z^2 \pi^2 \hbar^2}{2mL_z^2} + \frac{\hbar^2 k_x^2}{2m} + \frac{\hbar^2 k_y^2}{2m}.$$

Quantum confinement energy is along the z-axis and Kinetic Energy components along x and y direction

$p = \hbar k$

~~Based on the~~

Quantum wells are used to make semiconductor lasers, more efficient LEDs.



Quantum well 2D

[A quantum well laser is a laser diode in which the active region of the device is so narrow that quantum confinement occurs. The wavelength of the light emitted by a quantum well laser is determined by the thickness of the active region]

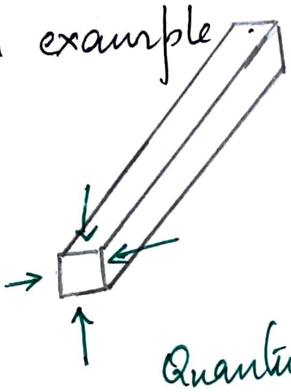
2. Nanowire (Quantum wire)

In quantum wire, quantum confinement is in two directions and only one direction is free for motion with any kinetic energy.

$$\psi_n(x,y,z) = \sqrt{\frac{2}{L_y}}\sqrt{\frac{2}{L_z}}\sin\left(\frac{n_y \pi y}{L_y}\right)\sin\left(\frac{n_z \pi z}{L_z}\right)e^{ik_x x}$$

$$E = \frac{\pi^2 \hbar^2}{2m}\left[\frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2}\right] + \frac{\hbar^2 k^2}{2m}$$

Quantum wires can be used to link or connect tiny components in nanocircuits. Motion of carriers in carbon nanotube is an example



Quantum wire 1D

3) Quantum dots

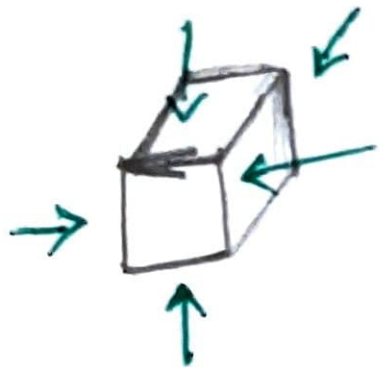
Quantum dots are zero dimensional structures in which the carriers are confined in all three dimensions. Their energy states are quantised in all three directions.

The corresponding wavefunction and energy is

$$\psi_n(x,y,z) = \sqrt{\frac{2}{L_x}}\sqrt{\frac{2}{L_y}}\sqrt{\frac{2}{L_z}} \sin\left(\frac{n_x \pi x}{L_x}\right) \sin\left(\frac{n_y \pi y}{L_y}\right) \sin\left(\frac{n_z \pi z}{L_z}\right)$$

$$E = \frac{\pi^2 \hbar^2}{2m}\left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2}\right)$$

Quantum dot makes ideal laser material. It can emit any colour from blue to red depending on the size of the dot.



Quantum Dot OD